

Ahmed Ayman

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Professional Summary

Fresh Graduate AI Engineer with strong hands-on experience in Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, and LLM-based systems. Skilled in building end-to-end AI solutions, including model training, evaluation, deployment, and optimization. Experienced in Python, TensorFlow, PyTorch, Scikit-learn, FastAPI, Docker, SQL, and cloud-based workflows. Passionate about solving real-world problems using AI and developing scalable, production-ready applications.

Technical Skills

- **Programming Languages:** Python, SQL, JavaScript
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost
- **Natural Language Processing:** Hugging Face Transformers, spaCy, NLTK, Text Classification, NER, Summarization, Prompt Engineering
- **Large Language Models & GenAI:** GPT, LLaMA, Gemini, RAG, LangChain, LangGraph, Vector Databases, Prompt Engineering
- **Computer Vision:** OpenCV, YOLO, CNNs, Image Classification, Object Detection
- **Deployment & MLOps:** FastAPI, Flask, Streamlit, Gradio, Docker, MLflow, REST APIs
- **Databases & Data Tools:** MySQL, PostgreSQL, MongoDB, Pandas, NumPy, Matplotlib
- **Vector Databases:** FAISS, Pinecone, Weaviate
- **Cloud & Tools:** AWS, GCP, Git, GitHub, Jupyter Notebook, Linux

Education

Bachelor's Degree in Computer Science

Cairo University, Cairo, Egypt | 2022 - 2026

Relevant Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Natural Language Processing, Computer Vision, Data Structures, Algorithms, Database Systems, Software Engineering, Cloud Computing

Graduation Project

AI Recruitment Intelligence System

2025 - 2026

- Designed and developed an AI-powered recruitment platform to analyze CVs and rank candidates against job descriptions.
- Built a hybrid evaluation pipeline combining deterministic scoring and LLM-based qualitative analysis.
- Implemented resume parsing, skill extraction, project-depth evaluation, education-fit scoring, and candidate ranking.
- Added support for GenAI evidence detection, prompt engineering signals, and deployment readiness.
- Improved ranking fairness by handling hallucination guards, missing data logic, and skill synonym mapping.
- **Technologies:** Python, FastAPI, LLM APIs, NLP, Scikit-learn, Pandas, Docker.

Projects

AI-Powered Recruitment Assistant

- Built a recruitment assistant that parses resumes, extracts structured candidate data, and matches applicants with job descriptions.
- Implemented scoring logic for technical skills, experience, projects, certifications, and education relevance.

- Added prompt-based LLM analysis to generate interview questions, strengths, and hiring recommendations.
- Deployed the system as an interactive web app for HR workflows.
- **Technologies:** Python, FastAPI, Streamlit, NLP, LLMs, Scikit-learn

RAG-Based Question Answering System

- Developed a Retrieval-Augmented Generation system for document-based question answering.
- Embedded documents using vector representations and stored them in FAISS/Pinecone for semantic retrieval.
- Integrated GPT/LLaMA-based answering pipeline with prompt engineering and context ranking.
- Improved response relevance and reduced hallucinations through chunking and retrieval filtering.
- **Technologies:** Python, LangChain, FAISS, Pinecone, GPT, LLaMA

NLP Pipeline for Text Classification and NER

- Built an NLP pipeline for text preprocessing, classification, and named entity recognition.
- Applied tokenization, feature extraction, transformer-based models, and evaluation metrics.
- Achieved strong performance on multi-class text classification tasks.
- **Technologies:** Python, Hugging Face, spaCy, NLTK, Scikit-learn

Brain Tumor Detection System

- Developed a CNN-based computer vision model for medical image classification.
- Trained and evaluated deep learning architectures for detection accuracy and generalization.
- Applied preprocessing, augmentation, and optimization techniques to improve model performance.
- **Technologies:** Python, TensorFlow, Keras, OpenCV

Real-Time Helmet Detection

- Implemented a real-time safety helmet detection system using YOLO and OpenCV.
- Built an inference pipeline for live video processing and detection visualization.
- Optimized detection speed and model accuracy for practical use.
- **Technologies:** Python, YOLO, OpenCV

Claims Data Analyzer

- Built an AI-assisted analytics tool for messy claims datasets with inconsistent schemas.
- Designed preprocessing pipelines for cleaning, validation, anomaly detection, and fraud flagging.
- Presented outputs through dashboards for decision support.
- **Technologies:** Python, Pandas, Pydantic, Streamlit

Internship Experience

AI Engineering Intern

Tech Company, Cairo, Egypt | Summer 2025

- Assisted in developing ML and DL models for real-world business use cases.
- Built preprocessing pipelines and contributed to training workflows for NLP and CV models.
- Worked with FastAPI APIs for serving model predictions.
- Helped containerize AI services using Docker.
- Collaborated with mentors to optimize model inference speed and deployment reliability.

Machine Learning Intern

AI Solutions Company, Cairo, Egypt | Summer 2024

- Developed ML models using Scikit-learn and PyTorch.

- Contributed to feature engineering, model evaluation, and experiment tracking.
- Worked with SQL databases and analytics pipelines.
- Supported an internal AI assistant prototype using prompt engineering and LLM APIs.

Certifications

- Machine Learning Specialization
- Deep Learning Specialization
- Natural Language Processing with Transformers
- Building LLM Applications with Prompt Engineering
- Docker for AI Deployment
- AWS Cloud Practitioner
- Generative AI Fundamentals

Key Achievements

- Built multiple end-to-end AI projects covering NLP, CV, LLMs, and deployment.
- Developed production-style APIs using FastAPI and Flask.
- Implemented RAG pipelines using FAISS and Pinecone.
- Worked on model deployment, Docker containerization, and MLflow experiment tracking.
- Strong alignment with AI Engineer role requirements in ML, DL, NLP, CV, and deployment.

Soft Skills

Problem Solving, Analytical Thinking, Team Collaboration, Communication, Research Mindset, Continuous Learning

Languages

Arabic: Native

English: Very Good